Project Plan: QIntern 2026

Title:
Ground-State Energy Estimation of spin-1/2 antiferromagnetic Heisenberg Model by Hamiltonian Calibration through Hardware-Efficient VQE

Duration: 6 weeks (1.5 months), *Team meetup twice a week*
Mentor: Dr. Muhammad Ahsan

Project Prerequisites:

Successful applicants are supposed to have a solid understanding of quantum mechanics (superposition, entanglement, quantum measurement) and to have some familiarity with spin systems and magnetic materials (Ising model). Proficiency in Python and prior experience with Qiskit 2.0 or an equivalent quantum SDK are required. Familiarity with IBM Quantum simulators and Runtime primitives (SamplerV2, EstimatorV2) and circuit transpilation is strongly preferred.

Project Overview

The spin-1/2 Kagome antiferromagnetic Heisenberg (KAFH) model is a prototypical frustrated quantum magnet predicted to host a quantum spin liquid ground state. Accurately estimating its ground-state energy on large 2D lattices remains a challenge for classical methods due to exponential complexity. Variational Quantum Eigensolvers (VQE) offer a hybrid quantum-classical path, but deep, problem-specific ansätze are difficult to simulate classically and hard to run on near-term hardware.

A recent approach (M. Ahsan, ACM Trans. On Q. Comp. 2026) introduces Hamiltonian calibration: instead of complicating the quantum circuit, the Hamiltonian itself is modified by locally increasing exchange couplings on selected defect triangles using a hardware-efficient ansatz (1D CNOT ladder). This enables accurate ground-state energy estimation even with limited entanglement resources.

The present project will implement and validate this Hamiltonian calibration strategy on Qiskit Aer and matrix product state (MPS) simulators. This provides a safe, reproducible, and educational entry point into utility-scale VQE for frustrated magnets.

Methodology (Step-by-Step Plan)

Week 1: Onboarding & Foundations
- Read and understand the related literature (focus on Hamiltonian calibration, defect triangles, ODR).
- Review spin-1/2 Heisenberg model, Kagome lattice geometry, and VQE basics.
- Set up Qiskit 1.0+ and run a simple VQE example on simulator to confirm environment.

Week 2: Small-Patch VQE (Original Hamiltonian)

- Implement 6-site Kagome patch, build Heisenberg Hamiltonian.
- Compute exact energy via exact diagonalisation.
- Construct hardware-efficient ansatz (CNOT ladder + R_y gates).
- Run noiseless VQE; record energy error (~10% as expected).

Week 3: Hamiltonian Calibration (J' Scan)

- Identify a defect triangle; increase one bond to J'.
- Scan J' from 1.0 to 2.5; find value minimising error to exact original energy (target ≈1.95).
- Repeat for 8-, 12-, and 19-site patches; verify same J' works.

Week 4: Local-Global VQE Decomposition

- Partition 19-site ansatz into two overlapping sub-circuits.
- Run local VQE (classical) on each sub-circuit; fix internal parameters.
- Reconstruct full ansatz with one $Ry(\theta)$ at junction; define reduced H_{SEL}.
- Optimise θ to minimise $\langle H_{SEL}\rangle$; then compute total $\langle H_{pert}\rangle$.
- Verify that total energy reaches the exact value (−29.14J).

Week 5: Noise Simulation & ODR Mitigation (Optional)

- Add depolarising noise (single-qubit error 0.001, CNOT error 0.01) to Aer simulator.
- Build reference Clifford ansatz (static dimer state $|\phi\rangle$); compute exact $\langle\phi|H|\phi\rangle$.
- Run reference circuit under noise → noisy $\mathrm{Tr}(\sigma H)$.
- Run global VQE circuit under noise → noisy $\mathrm{Tr}(\rho H_{pert})$.
- Apply ODR correction: mitigated$=\frac{\langle\phi\rangle}{Tr(\sigma H)}\times Tr(\rho H_{pert})$
- **Week 6: Bond Energy Maps & Final Reporting**
- Compute bond energies $O^{i,j}$ for all bonds under noise.
- Apply uniform ODR scaling to all bonds; plot difference map vs. classical MPS.
- Implement observable-wise ODR using two reference circuits; plot second map.

Deliverables:

All code and data will be released as open-source resources. Deliverables include the documented Jupyter notebooks and a written report in conference paper format suitable for arXiv submission. A final presentation to the QWorld's QIntern cohort will disseminate the findings.

References

- Ahsan, M. (2026). *Utility-scale Experimental Quantum Computation with Hardware Efficient Ansätze and Calibrated Hamiltonian* ACM Trans. On Q. Comp. 2026.
- Depenbrock, S. et al. (2012*). Nature of the spin-liquid ground state of the S=1/2 Heisenberg model on the Kagome lattice*. *Phys. Rev. Lett.* 109, 067201, 2012.